InvestWise

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AI-Driven Personalized Investment Portfolio Planning Platform for the CSE

SLIIT FACULTY OF COMPUTING

Supervisor Panel

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Introduction

Challenges Faced by Sri Lankan Investors:

- No single integrated platform providing all key investment insights.
- Investment information is scattered and fragmented across multiple sources.
- Lack of frequent, fine-grained, and explainable updates on company-specific and sector sentiment.
- Financial analysis tools often lack localization and focus on aggregate market trends.
- Investors struggle to combine sentiment, risk analysis, dividend forecasts, and price movement data effectively.

Research Question

How can AI-driven personalized investment platforms leverage advanced sentiment and market data analysis to optimize portfolio planning and improve returns on the Colombo Stock Exchange?

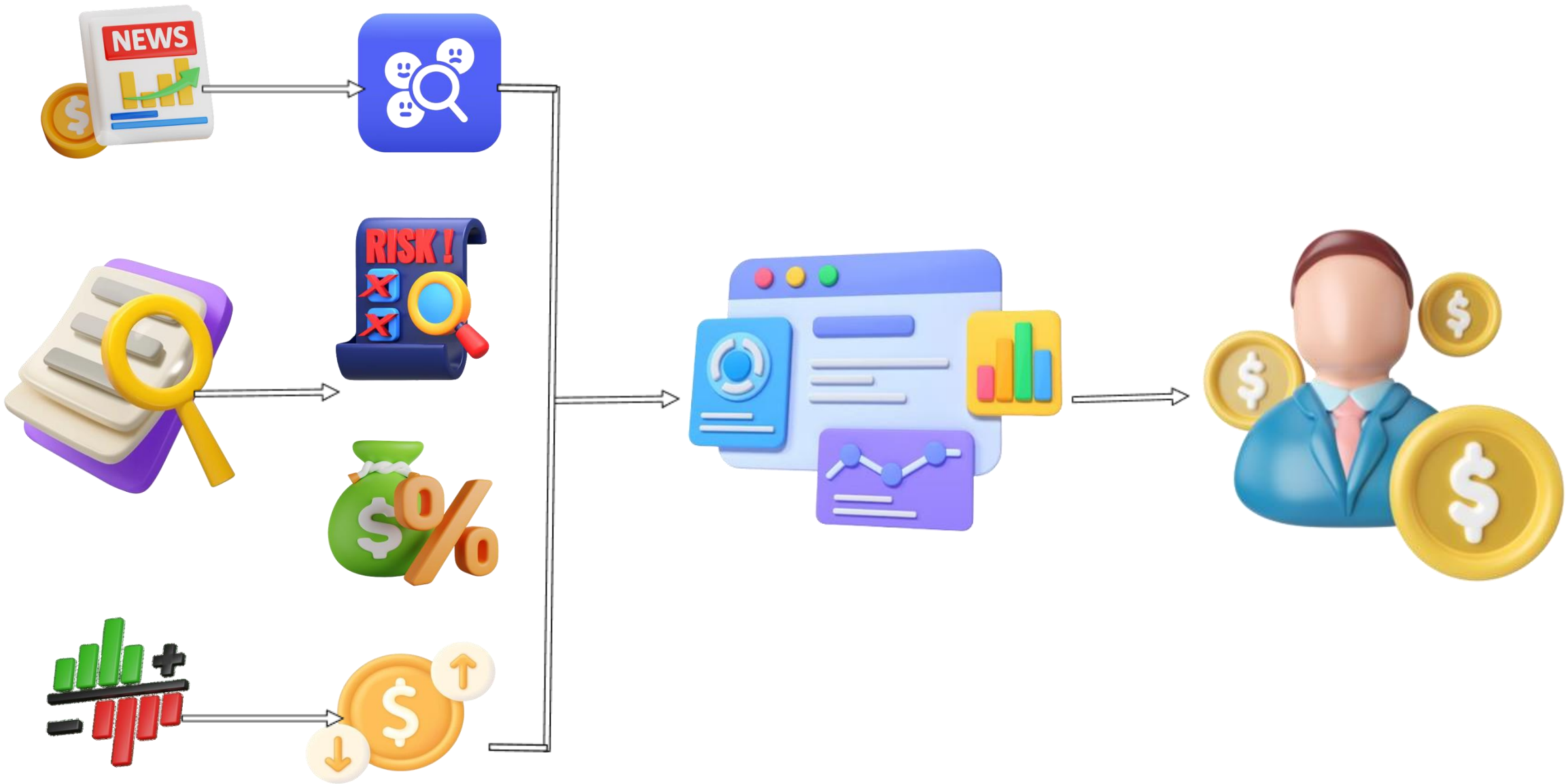
Objectives

Develop a user-friendly platform providing investment insights for CSE-listed companies.

Enable users to assess companies based on four key criteria:

- Sentiment analysis of companies based on financial news
- Financial company risk assessment
- Dividend prediction models
- Weekly stock price movement forecasts

System Diagram



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EVENT-AWARE SENTIMENT ANALYSIS USING ADVANCED NLP TO GAUGE MARKET SENTIMENT AROUND CSE COMPANIES

Software Engineering Specialization

Proven Gap/ Creative Solution

Proven Gap

- Limited Localized Solutions trained on SriLankan data
- Insufficient Integration of multi-source News Events
- Missing Company-level and Sector-level Sentiment Insights
- Infrequent Updates in a Single Consolidated Source

Creative Solution:

- Localized Event-Aware Sentiment Analysis Model
- Multi-Source News and Sentiment Integration
- Granular Sentiment Insights at Company & Sector Levels
- Frequent, Single-Source Updates

Primary Gap: Lack of a multi-level Sentiment analysis model exclusively trained on Sri Lankan data that provides timely updates on company/sector moving news

Current Shortcoming	Proposed Solution
Lack of NLP models fine-tuned on Sri Lankan financial news	NLP models fine-tuned exclusively on Sri Lankan financial news
Sentiment analysis at aggregate level	Classify specific market news per company and sector level
No timely updates	Timely updates

Key Domains Utilized

Natural Language Processing & Deep Learning

- Use of transformer-based models (BERT, RoBERTa variants) fine-tuned on Sri Lankan financial news and disclosures to detect sentiment and classify event types (e.g., earnings, regulatory changes)
- Fine-grained Named Entity Recognition (NER) to identify CSE-listed companies, sectors, and event attributes with high precision.

Sri Lankan Stock Market Concepts(CSE)

- Open-price formation influenced by overnight global/local events
- Handling market complexities like corporate actions (splits, dividends) impacting sentiment and market reaction.

Data Engineering & DevOps

- Automated web-scraping pipeline with daily updates
- Scalable deployment via Docker, CI/CD pipelines

Latest Technologies Utilized

Transformer-based models (BERT variants) fine-tuned on local financial texts

Named Entity Recognition (NER) and event detection algorithms

Scalable cloud infrastructure for real-time data ingestion & model retraining

Data Availability

Public news portals & official CSE disclosures.

Price Movements from CSE Data Library

Twitter(X) posts from trusted/verified Twitter Handles

Evaluation & Validation

Model Performance: Cross-validation using 20% of the data set

Correlation analysis between sentiment scores and CSE stock movements

Conduct user testing with investors for alert relevance, interpretability, and decision usefulness.

Ethical Clearance

Ethical data usage adhering to platform terms & local privacy laws

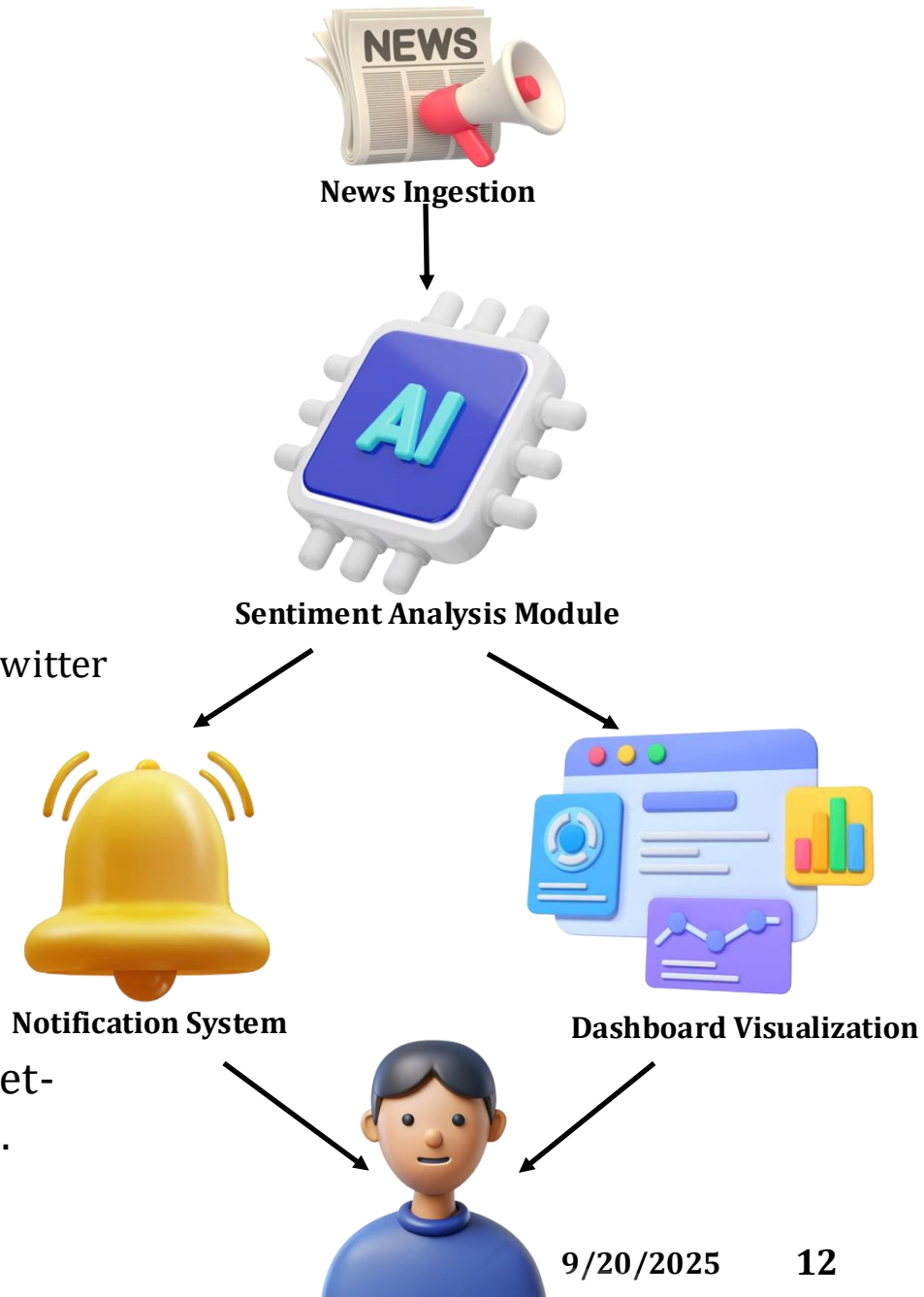
Transparency in AI-driven decisions for user trust

Core Architecture

Presentation Tier (Frontend / User Interface)
- Investor Dashboard
- Shows real-time sentiment classification (Positive / Negative / Neutral)
- Displays alerts and visual summaries of market sentiment trends

Application Tier (Business Logic)
- Pre-trained NLP Sentiment Analysis Model
- Classifies incoming financial news and social media into sentiment categories
- Triggers notifications based on significant sentiment changes

Data Tier (Data Management)
- Web scraping pipeline or API connectors to financial news sites, CSE disclosures, Twitter feeds
- Stores latest and historical text data and sentiment results
- Performs preprocessing: text cleaning, entity recognition, event tagging



Real-World Usage Scenario

Investors receive alerts and dashboard insights on positive or negative market-moving events, enabling faster, data-driven decisions ahead of price changes.

IT22306654 | ARIYARATHNE W D N

Software Engineering Specialization

Proven Gap/ Creative Solution

Proven Gap	Creative Solution
Few models predict CSE open prices; most focus on closing prices	Trend-aware LSTM model for open price prediction
Models rarely use momentum or trend-sensitive features	Use momentum features to boost forecast accuracy
Lack of automated, real-time forecasting and dashboards for investors	Automated pipeline and real-time investor dashboard
Raw CSE data is often noisy or incomplete	Robust data cleaning and event adjustment in preprocessing

Primary Gap: Lack of an accurate, real-time next-week open price forecasting model for Colombo Stock Exchange stocks that integrates momentum-based trends and operates within an automated, investor-focused platform.

Addressing Existing System Shortcomings

Current Shortcoming	Proposed Solution
Misses volatility and trend changes	Use trend-aware LSTM to capture patterns
Most tools focus on closing prices	Predict next-week open prices for CSE stocks
Models use only **raw OHLCV** (weak trend detection).	Add momentum for better trend detection
No automation: manual/lagged updates.	Automated **data pipeline** with daily ingestion

Key Domains Utilized

Time-Series Forecasting & Deep Learning

- LSTM model for next-week open price prediction
- Sequences use 5–10 day look-back for multi-step forecasting
- Combines OHLCV and momentum features for better trend prediction

Machine Learning & AI:

- Advanced deep learning with LSTM for multi-step prediction
- Feature engineering: momentum indicators for trend sensitivity
- Model validation using metrics like MAPE and RMSE

Data Engineering & Automation

- Automated scraping pipeline updates market data daily
- Robust preprocessing to engineer OHLCV + momentum features
- Stores processed data for use in forecasting and dashboard display

Latest technologies and evaluation

Latest Technologies Utilized

LSTM for open price forecasting

Momentum feature (5-day % change)

Automated daily data pipeline

Baselines: ARIMA, Prophet, XGBoost

Investor dashboard

How to Evaluate / Prove Success

Walk-forward validation across regimes

Ablation study: compare models with vs. without momentum

Data freshness, retraining logs, API latency

Target MAPE ≤ 8% for liquid stocks, >10% over baseline

Usability testing: KPIs (MAPE, RMSE), forecasts, alerts

Data Availability & Ethical Clearance

Data Availability

Official CSE Data Library purchased

Daily updates via automated pipeline.

Ethical Clearance

Only market data used – no personal/PII

No ethics approval needed

Data under CSE license terms, no redistribution

Real-world usage scenario

- **Plan Investments:** See predicted open prices for next week to decide when to buy or sell.
- **Get Instant Alerts:** Get notified when new forecasts or updates are ready.
- **Manage Risk:** Use predictions to adjust your investment risk.
- **Track Portfolio:** Monitor trends and key stats to make better decisions over time.

Core Architecture

1. Presentation Tier (Frontend / User Interface)

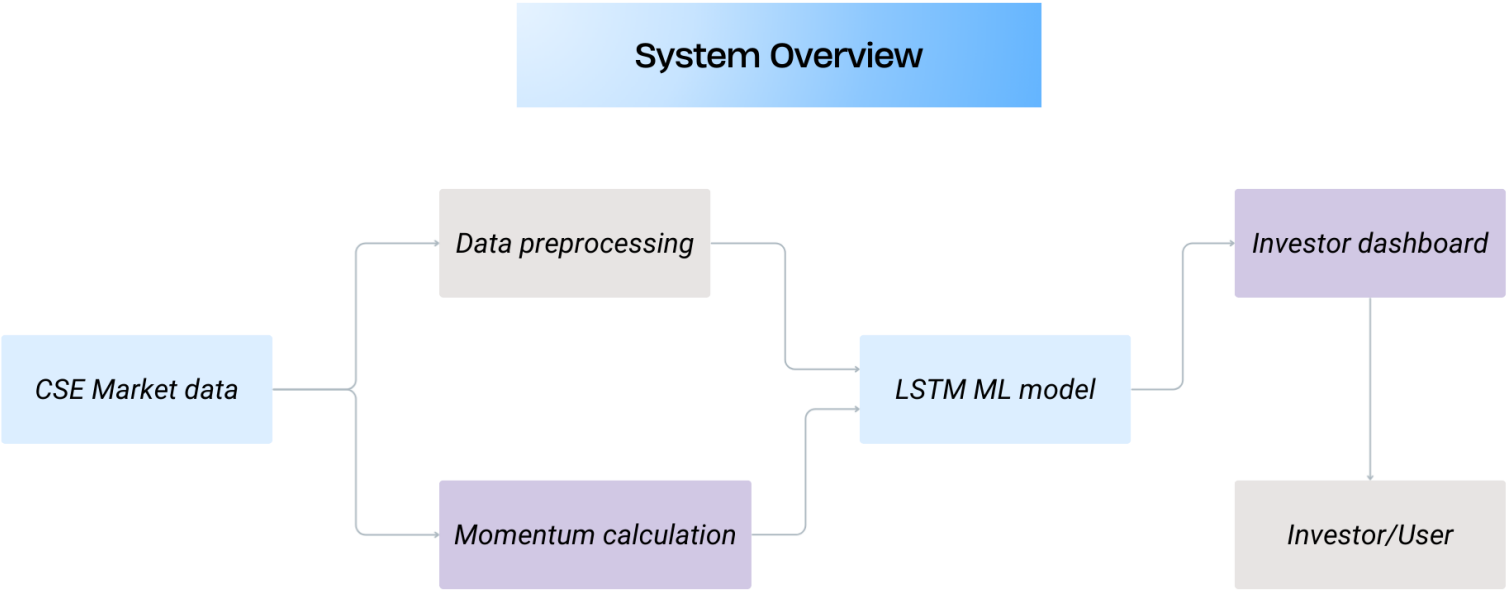
- Investor Dashboard
- Visualizes next-week open price predictions
- Shows charts, KPIs, and alerts

2. Application Tier (Business Logic)

- Pre-trained LSTM Model for prediction
- Uses OHLCV + momentum features
- Generates next-week open price forecasts

3. Data Tier (Data Management)

- Web-scraping pipeline for CSE market data
- Stores historical + latest data
- Handles preprocessing: cleaning, missing values, feature engineering



IT22271532 | H.M. MADHAWA AWISHKA

Comprehensive Financial Health Analysis Module for CSE Listed Companies

Software Engineering Specialization

Proven Gap and Creative Solution

Current Problem

- CSE lacks sophisticated AI-powered financial risk assessment tools
- International platforms don't account for Sri Lankan market conditions
- Manual analysis methods are time-consuming and error-prone
- Lack of transparency in existing risk assessment systems.

Creative Solution:

- AI-powered financial health analysis module specifically for CSE-listed companies
- Explainable AI (XAI) integration for transparent risk predictions
- Automated data processing from CSE reports
- Machine learning models (Random Forest, XGBoost) for accurate risk assessment

Primary Gap: Absence of any CSE-focused platform combining advanced machine learning models and explainable AI techniques for transparent, accurate financial risk assessment.

Addressing Existing System Shortcomings

Current Shortcoming	Proposed Solution
Black Box prediction without explanation	XAI techniques with actionable insights
Generic International tool	CSE-specific algorithms trained on local data
Simple Statical methods	Adavanced ML algorithms for pattern recognition

Key Domains Utilized

Financial Risk Management:

- Financial ratio analysis (liquidity, leverage, profitability, efficiency ratios)
- CSE market dynamics and regulatory requirements

Machine Learning & AI

- Advanced algorithms: Random Forest, XGBoost
- Financial data preprocessing techniques
- Model validation and performance evaluation

Explainable AI

- SHAP and LIME implementation for transparent predictions
- Interpretability frameworks for stakeholders

Software Engineering and DevOps

- Containerization with Docker for deployment consistency
- Kubernetes orchestration for scalable microservices architecture
- CI/CD pipelines for automated testing and deployment

Latest Technologies

- Explainable AI (SHAP, LIME) for transparent risk-factor attribution
- Random Forest / XGBoost: ML algorithms for financial risk classification
- Automated, real-time ingestion of CSE financial statements and market data

Success Validation Methods

- Model performance: cross-validation, backtesting, precision
- Explainability testing: SHAP consistency and expert review
- Processing speed: real-time analysis capability

Data Availability & Ethical Clearance

Data Sources:

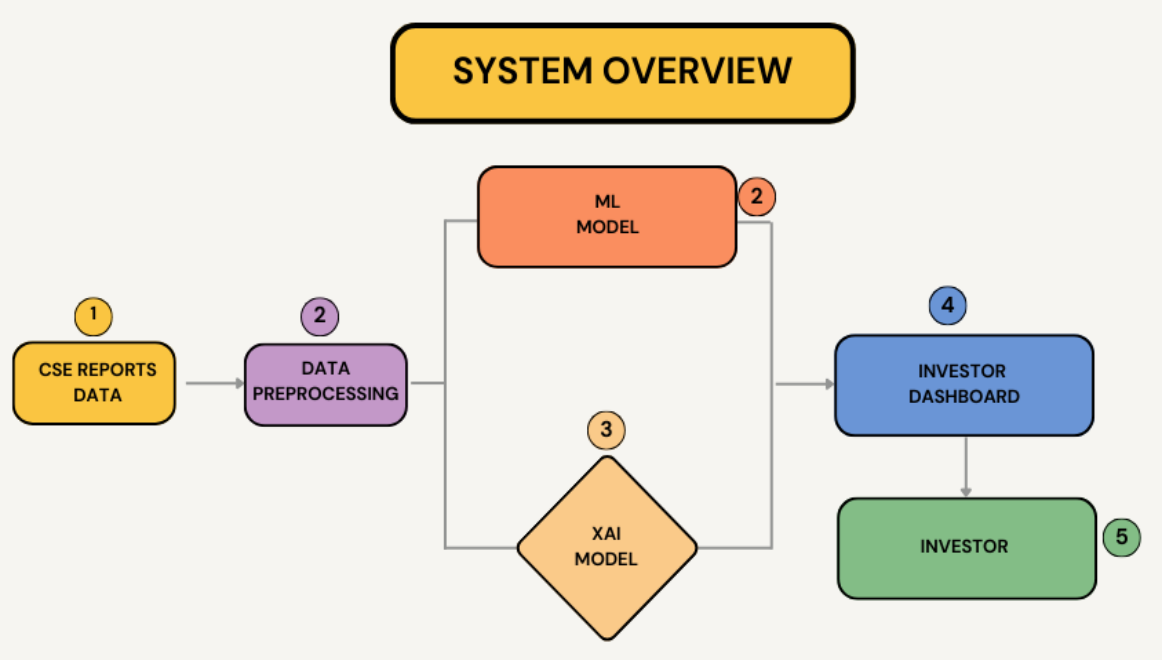
- Official CSE filings (annual, quarterly)
- ≥ 10 years of historical statements for companies

Ethical Compliance:

- Public data usage (no privacy concerns)
- Regulatory compliance with Sri Lankan securities law
- Bias mitigation and fairness metrics implementation
- Transparent algorithmic decision-making

Core Architecture

- Data Collection Layer: Automated extraction of financial statements (balance sheets, income statements, cash flows) from CSE official reports
- Data Processing Engine: Comprehensive preprocessing including data cleaning, standardization, and financial ratio calculations
- Machine Learning Module: Trained Random Forest and XGBoost models for financial risk prediction
- Explainable AI Layer: SHAP and LIME implementation for transparent risk explanations
- User Interface: Dashboard for risk analysis visualization and company comparisons



Real-World Usage Scenario

- Portfolio managers or user can select CSE-listed companies for risk assessment
- System will be giving financial risk analysis's for the specific company.
- Provides explainable actionable insights
- Enables comparative analysis across multiple companies for informed investment decisions
- Supports regulatory compliance through transparent, auditable AI explanations.

Student Must add a professional photo to this cage

IT22102614 | WAHALATHANTHRI DBD

Software Engineering Specialization

Proven Gap and Creative Solution

Problem

• research only analyzes dividend determinants.
•Investors manually review firm reports to get insights
•Lack of decision support tools
•Global Research single stage
•They focus on specific stock exchanges

Creative Solution:

- **Decision-support tool** for dividend forecasting in the Sri Lankan market delivered through a web-based dashboard with a two-stage machine learning framework

Primary Gap: In Srilankan context current research only explains dividend determinants without providing predictive or decision-support tools.

Addressing Existing System Shortcomings

Current Shortcoming	Proposed Solution
Research focus only on descriptive statistics	Introduces a ML to forecast dividends
Lack practical decision-support tools for investors .	Delivers an interactive dashboard with historical data + forecasts.
Only address pay or no pay question	Address both pay or not and amount as a two stage pipeline

Application of Specialization-Based Knowledge

Software Engineering Knowledge

- **System Architecture design :** layerd+microservices
- **Development Practices:** modular coding, testing,version control.
- **Deployment:** Containerization with Docker for portability and scalability.

Data Science Knowledge

- **Machine Learning Framework:** Two-stage ML (classification,regression)
- **Preprocessing:** lagged features, winsorization to handle ouliers
- **Feature Engineering:** Ratios, lagged indicators, macroeconomic features.
- **Evaluation:** Classification(Accuracy, Recall, ROC-AUC); Regression (RMSE, MAE, R^2).

Technologies

•**Scikit-learn** (baseline models, preprocessing).
•**XGBoost / LightGBM / CatBoost** (tree-based models for classification & regression).

Evaluation, Data & Ethics

Model Evaluation

- *Classification :* Accuracy, Precision, Recall, F1-score, ROC-AUC.
- *Regression :* RMSE, MAE, R^2.
- Benchmark against different models.

Success Criteria:

- Accurate predictions.
- reliable performance.
- User friendly interface.

System Performance

- Ability to handle multi-year datasets without errors.
- Prediction response should be fast and reliable under concurrentuser requests.
- Stable deployment across environments (local/cloud).

Data Availability

- **Firm Financials & Dividends:** Colombo Stock Exchange (CSE) reports.
- **Macroeconomics:** Central Bank of Sri Lanka, World Bank
- **Coverage:** At least 10 years of public historical data.

Ethical Clearance

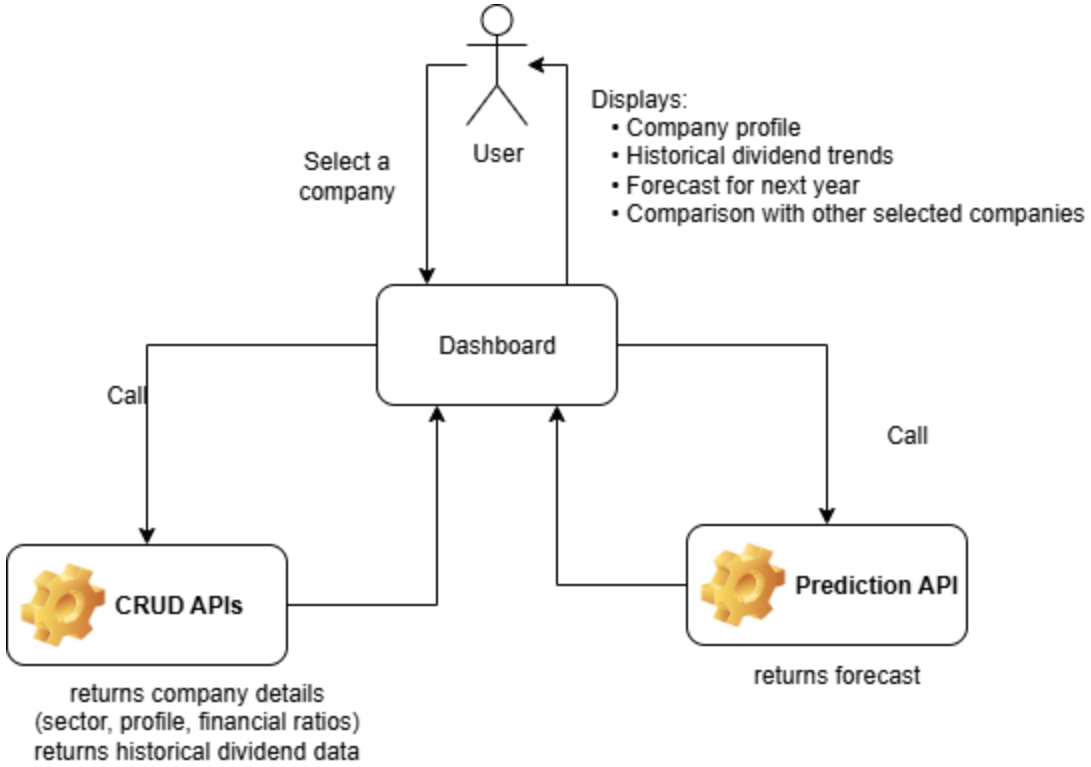
- Uses only **publicly available data**.
- No confidential or personal information involved.

Core Architecture

Layerd + Microservices (Hybrid)

- **Data Layer**
  - Historical dividends, firm-level financials, and macroeconomic indicators.
  - Stored in a structured database for consistency.
- **Model Layer**
  - Model artifacts stored and loaded for predictions.
- **API Layer**
  - **Prediction API:** Provides dividend forecasts.
  - **Company API:** Fetches company details (name, sector, financial ratios).
  - **History API:** Delivers past dividend records.
  - **Macro API:** Supplies macroeconomic data.
- **Presentation Layer**
  - Web dashboard built with React.js + Plotly/Recharts.
  - Features: company selection, historical trends, forecasts, and cross-company comparisons.

Real-World Usage Scenario



COMMERCIALIZATION

Ability of Commercialization

- Potential to extend beyond research into a market-ready financial decision-support tool
- High relevance for investors in emerging markets like Sri Lanka

Investment / Cost & Recovery

- Estimated costs: data acquisition, model training, system development, and maintenance
- Recovery: Subscription model, consultancy packages, and partnerships with brokerage firms

Target Market & Customer Profile

- Retail investors, investment advisors, and academic/industry researchers
- Customers seeking AI-driven insights to improve portfolio planning

Value Proposition

- Combines sentiment analysis, risk analysis, dividend prediction, and stock price forecasting
- Provides transparent, data-driven decision support for investment strategies

Intellectual Property Rights (IPR)

- Research-driven predictive algorithms and unique system design
- Opportunity for patenting models/methodologies and securing brand identity

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